# Wi-Fi Password Security Auditor

```
import getpass
import re

def wifi_password_auditor(password):
    score = 0
    suggestions = []

    print("\\n" + "=" * 48)
    print("     WI-FI PASSWORD SECURITY AUDITOR")
    print("=" * 48)

    if len(password) >= 16:
        score += 2
        print("[PASS] Excellent password length")
    elif len(password) >= 12:
        score += 1
        print("[PASS] Good password length")
    else:
        print("[FAIL] Password is too short")
        suggestions.append("Use at least 16 characters")

    checks = [
        (bool(re.search(r"[A-Z]", password)), "Uppercase letters detected", "Include uppercase
letters"),
        (bool(re.search(r"[a-z]", password)), "Lowercase letters detected", "Include lowercase
letters"),
        (bool(re.search(r"\\d", password)), "Numbers detected", "Include numbers"),
        (bool(re.search(r"[^A-Za-z0-9]", password)), "Special characters detected", "Include
special characters"),
    ]
    for passed, success, suggestion in checks:
        if passed:
            score += 1
            print("[PASS]", success)
        else:
            print("[FAIL]", suggestion)
            suggestions.append(suggestion)

    common = ["password", "12345678", "qwerty", "admin", "wifi123", "iloveyou"]
    if any(item in password.lower() for item in common):
        score = max(0, score - 2)
        print("[WARN] Common password pattern detected")
        suggestions.append("Avoid common words and sequences")
    else:
        print("[PASS] No listed common patterns detected")

    if re.search(r"(.)\\1{3,}", password):
        score = max(0, score - 1)
        print("[WARN] Repeated characters detected")
        suggestions.append("Avoid repeating the same character")
    else:
        print("[PASS] No long repeated-character pattern")

    print("\\n" + "-" * 48)
    print("SECURITY SCORE :", f"{score}/7")

    if score >= 7 and len(password) >= 16:
        print("SECURITY LEVEL : VERY STRONG")
        print("AUDIT STATUS   : EXCELLENT")
    elif score >= 5:
        print("SECURITY LEVEL : STRONG")
        print("AUDIT STATUS   : GOOD")
    elif score >= 3:
        print("SECURITY LEVEL : MODERATE")
        print("AUDIT STATUS   : NEEDS IMPROVEMENT")
    else:
        print("SECURITY LEVEL : WEAK")
        print("AUDIT STATUS   : ACTION REQUIRED")

    if suggestions:
        print("\\nSECURITY RECOMMENDATIONS:")
```

```
        for number, item in enumerate(suggestions, 1):
            print(f"{number}. {item}")
    else:
        print("\\nAll basic password checks passed!")

    print("\\nNOTE: This checks password characteristics only.")
    print("It does not connect to or test your actual Wi-Fi network.")
    print("=" * 48)

if __name__ == "__main__":
    print("=" * 48)
    print("       WI-FI PASSWORD SECURITY TOOL")
    print("=" * 48)
    password = getpass.getpass("Enter Wi-Fi password: ")
    wifi_password_auditor(password)
```